

State Board of Technical Education and Training, Andhra Pradesh
Diploma in Computer Engineering - C-23 Curriculum
CM-304: Data Structures through C - Mid-Term Examination (Unit Test-II)
Time: 90 Minutes Total Marks: 40

Part A **(16 Marks)**

Instructions: Answer all questions. Q1 carries 4 marks (1 mark each). Q2 to Q5 carry 3 marks each.

Q1. Answer the following: (4 marks)

- (a) Define a stack data structure. (1 mark)
- (b) What is underflow in a queue? (1 mark)
- (c) Name two applications of queues. (1 mark)
- (d) What is a priority queue? (1 mark)

Q2. Explain push and pop operations on a stack with examples. (3 marks)

Q3. Write the algorithm for evaluating a postfix expression using a stack. (3 marks)

Q4. Describe enqueue and dequeue operations in a queue. (3 marks)

Q5. Explain the concept of a circular queue. (3 marks)

Part B **(24 Marks)**

Instructions: Answer all questions. Each question carries 8 marks and has an internal choice.

Q6.

- (a) Write a C program to implement a stack using linked list with push, pop, and display operations. (8 marks)

OR

- (b) Explain the application of stacks in recursion with an example. (8 marks)

Q7.

- (a) Write a C program to implement a queue using an array with enqueue, dequeue, and overflow/underflow checks. (8 marks)

OR

- (b) Describe the differences between linear queue and circular queue with advantages. (8 marks)

Q8.

- (a) Write a C program to implement a priority queue using arrays. (8 marks)

OR

- (b) Explain deque (double-ended queue) operations and its types. (8 marks)